

Course Syllabus - Fall A 2026

CSE 551: Foundations of Algorithms

Course Description

Algorithms, or a step-by-step process to efficiently reach the desired goal, have been part of human history since the 1200s. Algorithms are a fundamental component to any computerized system. In this foundational course, you will learn several different algorithms and be able to explain how they work and why they are considered good. This knowledge will help you:

- Evaluate appropriate algorithmic techniques that can lead to more efficient solutions for a problem, instead of just coding the first idea that comes to mind.
- Develop sound background knowledge on algorithms that will allow you to navigate the field's literature, beyond the context of this class.

In order to gain these skills, you will have to work through and understand several algorithmic techniques, including the mathematics necessary for analyzing the properties of these techniques and the algorithms based on them.

Specific topics covered include:

- Greedy Algorithms
- Stable Matching
- Divide-and-Conquer
- Dynamic Programming
- Amortized Analysis
- Network Flows
- NP-completeness
- Introduction to Randomized and Approximation Algorithms

Learning Outcomes

Learners completing this course will be able to:

CSE 551 Syllabus
Fall A 2026

Any information in this syllabus may be subject to change without advance notice. Stay informed by checking course announcements and the syllabus section of your course.

- Identify and apply algorithmic techniques to solve a problem.
- Apply knowledge of algorithms in multiple contexts using multiple programming languages.
- Evaluate correctness and efficiencies of algorithms.

Estimated Workload/ Time Commitment Per Week

Average of 18 - 20 hours per week

Required Prior Knowledge and Skills

This course will be very challenging, and learners are expected to learn the necessary technologies on their own time.

Proficient Mathematical Skills and Theoretical Understanding

- Asymptotic Notation (Big-Oh)
- Recurrence relations
- Mathematical proofs
- Recursion
- Basic discrete math (e.g., sets, functions, logic, graphs, etc.)
- Worst-Case Analysis

Strong Application Skills

- Python is required for coding assignments.
- Any one of the well-known programming languages like: C, C++, Java or Python

Proficient Experience

- Data structures and algorithms such as Sorting Algorithms, Hash Tables, Binary Search Trees, Heaps, and Red-Black Trees
- Greedy Algorithms, Divide-and-Conquer, Dynamic Programming
- Graph Algorithms such as Depth-First Search, Breadth-First Search, Minimum Spanning Trees (Kruskal's and Prim's Algorithms), and Shortest-Paths (Dijkstra's Algorithm)

CSE 551 Syllabus

Fall A 2026

Any information in this syllabus may be subject to change without advance notice. Stay informed by checking course announcements and the syllabus section of your course.

Course Access

Your ASU courses can be accessed by both my.asu.edu and asu.instructure.com; bookmark both in the event that one site is down.

Technology Requirements

Proctoring

- [LockDown Browser Computer Requirements for Installation](#)
- [Honorlock Minimum System Requirements](#)

Hardware

- Standard personal computer with major operating system
- Reliable, strong Internet connection
- Webcam
- Microphone

Software/Other

- Current web browser
- EdDiscussion

Technology Recommendations

N/A

Textbook and Readings

At the graduate level, inquiry, research, and critical reading are part of the learning experience; however, this course does not have a required textbook.

For interested learners, the faculty course designer recommends this text:

CSE 551 Syllabus

Fall A 2026

Any information in this syllabus may be subject to change without advance notice. Stay informed by checking course announcements and the syllabus section of your course.

J. Kleinberg and E. Tardos, *Algorithm Design*. United States: Pearson Education, 2006.

Course Schedule and Important Dates

Course teams will not be working on ASU's days off* and those are listed in the Course Schedule. Please review the [ASU Days Off](#) for more details.

Module: Title	Begins at 12:01 AM Arizona (AZ) Time	Ends at 11:59 PM Arizona (AZ) Time
Orientation and Onboarding Review You must complete required tasks in the Orientation and Onboarding Review for Module 0: Welcome and Start Here to be unlocked.	August 17, 2026	August 19, 2026
Module 0: Welcome and Start Here You must complete required tasks in Module 0: Welcome and Start Here for the rest of the course to be unlocked.	August 17, 2026	August 20, 2026
Module 1: Stable Matching	August 17, 2026	August 23, 2026
Module 2: Greedy Algorithms	August 24, 2026	August 30, 2026
Module 3: Amortized Analysis and Splay Trees	August 31, 2026	September 6, 2026
Module 4: Divide and Conquer ASU Holiday: Labor Day, September 7, 2026	September 6, 2026	September 13, 2026
Exam 1	September 14, 2026	September 20, 2026
Module 5: Dynamic Programming	September 14, 2026	September 20, 2026
Module 6: Network Flows	September 20, 2026	September 27, 2026

CSE 551 Syllabus Fall A 2026

Any information in this syllabus may be subject to change without advance notice. Stay informed by checking course announcements and the syllabus section of your course.

Course Evaluation You may also refer to ASU's Office of Evaluation and Educational Effectiveness (UOEEE) for dates.	September 21, 2026	October 4, 2026
Module 7: Polynomial Time Reductions and NP-Completeness	September 28, 2026	October 4, 2026
Module 8: Approximation and Randomized Algorithms	October 4, 2026	October 9, 2026
Exam 2	October 4, 2026	October 9, 2026
Deadline to Submit all Graded Coursework	N/A	October 9, 2026
Course Closes Past this date, Ed Discussion will no longer be monitored. Please download copies of what you would like from the course.		November 24, 2026

Grades are due Monday, October 12, 2026. Please see the [for additional information.](#) [ASU Academic Calendar](#)

Late or Missed Coursework

When possible, kindly notify the instructor before a coursework deadline by creating a private thread in Ed.

If an urgent situation or emergency arises and you are unable to submit the assignment on time, please send the instructor a private thread on Ed as soon as you are able to.

Follow the appropriate University policies to request an [accommodation for religious practices](#) or to accommodate a missed assignment [due to University-sanctioned activities](#).

CSE 551 Syllabus
Fall A 2026

Any information in this syllabus may be subject to change without advance notice. Stay informed by checking course announcements and the syllabus section of your course.

Coursework Due Dates and Late Penalties

Unless otherwise noted, all coursework is due on **Sundays at 11:59 PM Arizona (AZ) time**. Due dates in your course are set up in Arizona Standard time. Use a [Time Converter](#) to ensure you account for the difference in time zones and remember to update your course settings to reflect your time zone (see your onboarding course for directions). Arizona does **not** observe daylight savings time.

Review specific due dates directly in your course. For learners with accommodations through [Student Accessibility and Inclusive Learning Services \(SAILS\)](#) and/or the [Pat Tillman Veterans Center \(PTVC\)](#), please work with your SAILS consultant and/or PTVC Advocacy Team, Connect, and your instructor.

Module Quizzes

No late penalty is applied after the scheduled due date and time, but in order to complete all work by the final submission date, it is highly recommended to follow the course schedule.

- **Module 1 Graded Quiz**
- **Module 2 Graded Quiz**
- **Module 3 Graded Quiz**
- **Module 4 Graded Quiz**
- **Module 5 Graded Quiz**
- **Module 6 Graded Quiz**
- **Module 7 Graded Quiz**
- **Module 8 Graded Quiz**

Homework Quizzes

- **Assignment Quiz 1 Graded Quiz**
- **Assignment Quiz 2 Graded Quiz**
- **Assignment Quiz 3 Graded Quiz**

CSE 551 Syllabus
Fall A 2026

Any information in this syllabus may be subject to change without advance notice. Stay informed by checking course announcements and the syllabus section of your course.

- **Assignment Quiz 4 Graded Quiz**
- **Assignment Quiz 5 Graded Quiz Part 1 and Part 2**
- **Assignment Quiz 6 Graded Quiz**
- **Assignment Quiz 7 Graded Quiz**
- **Assignment Quiz 8 Graded Quiz**

Coding Assignment

- **Module 1 Coding Assignment**
- **Module 2 Coding Assignment**
- **Module 3 Coding Assignment**
- **Module 4 Coding Assignment**
- **Module 5 Coding Assignment**
- **Module 6 Coding Assignment**
- **Module 7 Coding Assignment**
- **Module 8 Coding Assignment**

Exams

An automatic late penalty of 100% is applied after the scheduled due date and time. All practice and required coursework should be completed prior to taking your exams.

- **Exam 1**
- **Exam 2**

Course Content

Each course in the MCS program is uniquely designed by expert faculty, so learners can best master the learning outcomes. As a result, course features and experiences are not the same across all MCS courses. Learners are expected to plan accordingly to accommodate for these differences.

CSE 551 Syllabus
Fall A 2026

Any information in this syllabus may be subject to change without advance notice. Stay informed by checking course announcements and the syllabus section of your course.

Content and Assessment Details

If you have specific questions related to instructional and assessment items in this course that you would like to be considered to be addressed in the Zoom meeting hosted by the instructor, please clearly indicate your request in your Ed Discussion thread.

For details regarding how points are earned in different types of assessments, please refer to your onboarding courses.

WeVideo Lecture Playlists

The course content is presented through a collection of WeVideo Playlists embedded in each module. WeVideo is a video platform that prompts interaction and note-taking while viewing course content. The playlists launch automatically and you can playback the course content by selecting the video titles in the playlist. The videos can be rewatched, but playlist videos cannot be downloaded. The playlist pages will include the downloadable video transcripts and any applicable supplementary material. Other course materials that accompany the lectures will be found in the media guides.

A media guide is included at the beginning of each module in the Overview section. These guides are designed to give you a snapshot description of each module's media components and to provide PDF lecture slides or note-taking materials where available, so you can plan your learning and quickly go back and review material as you prepare for your coursework.

WeVideo Interactions: In-Video Questions and Knowledge Checks

WeVideo provides opportunities for interaction and reflection as you learn the course content. During lecture videos, there are interactions referred to as in-video questions. After videos, there are interactions called knowledge checks. Interactions are designed to support your learning, highlight specific content, encourage active viewing and/or note-taking, and provide practice opportunities. They are untimed, ungraded learning opportunities to test your knowledge of the concepts presented during the lecture videos. You may retake these as often as you would like at any point in the course.

You can toggle the clipboard icon on the left of the screen and select a review to see all the questions. You are accountable for this information as it may be assessed in different ways in other graded coursework.

There are no late penalties. Interactions are **not** counted toward your final grade in the class.

CSE 551 Syllabus Fall A 2026

Any information in this syllabus may be subject to change without advance notice. Stay informed by checking course announcements and the syllabus section of your course.

Discussions

Ed Discussion

Ed Discussion (Ed) is being used in place of Canvas Discussion Forums. The purpose of Ed Discussion is to provide a place for learners to ask questions and receive answers from course staff and peers about course content and coursework. The course team is engaged in discussions, but it is also a space to clarify, support, and enrich learner-to-learner communication and learning. There are designated categories for course items. You must select a category and subcategory to start a thread.

Discussions in Ed are designed to provide:

- Clarification
- Feedback
- Enrichment and deeper learning
- Connections between concepts or key ideas
- Reflection opportunities of real-world experiences
- Respectful debate and perspective building
- Resource sharing
- Networking

There are no late penalties. Ed is **not** counted toward your final grade in the course.

Designated Assignment Discussion in Ed Discussion

Use Ed to discuss items relating to the course assignments. Questions/Threads should be categorized by their designated title in Ed. Please check for questions already asked and answered, or marked as resolved.

There are no late penalties. Responses in Ed are **not** counted toward your final grade in the course.

Assignments

Modules 1-4, 6-8 each include one (1) assignment. Module 5 has two (2) assignments. There is a total of 9 assignments. Each assignment consists of a downloadable document that presents a problem (or multiple problems) and typically includes four - six (4-6) questions. You are encouraged to download these documents and fill them out. Your answers are not submitted for grading, and you will not receive a score. Assignments prepare you for the exams.

CSE 551 Syllabus

Fall A 2026

Any information in this syllabus may be subject to change without advance notice. Stay informed by checking course announcements and the syllabus section of your course.

All assignment documents and materials are provided in *Module 0: Welcome and Start Here* of your course, so you can preview what is expected and design your own learning schedules to complete these on time.

Each assignment has accompanying solutions provided to you so you can learn from your mistakes and reinforce your correct thinking. These are for your personal reference while you are a learner in this course. They are not to be shared with anyone else in any way outside of this per ASU's academic integrity and copyright policies. Violations of this agreement will be strictly enforced.

There are no late penalties. Assignments are **not** counted toward your final grade in the class.

Assignment Quizzes

There are EIGHT (8) Assignment Quizzes in this course—one tied to each Assignment.

Take each Assignment Quiz after you complete the paired Assignment. Solution documents are sequenced to unlock only after the associated quiz is completed. Use these unlocked solutions as formative study aids by checking your reasoning, comparing approaches, and targeting your review; they're for learning support, not for copying into graded work.

Proctored Exams

You have TWO (2) proctored exams. These consist of Exam 1, Exam 2. The proctored exams do not include feedback. Read the Graded Quiz and Exam Policy for your course for more information.

No late exams will be permitted or accepted and will result in a score of zero (0) points. This does not include established accommodations for learners receiving accommodations through [Student Accessibility and Inclusive Learning Services \(SAILS\)](#) and and/or the [Pat Tillman Veterans Center \(PTVC\)](#).

Proctored exams count toward your final grade in the class.

Exam Details	Exam 1	Exam 2
Content Covered	Modules 1, 2, 3, and 4	Modules 5, 6, 7, and 8

CSE 551 Syllabus
Fall A 2026

Any information in this syllabus may be subject to change without advance notice. Stay informed by checking course announcements and the syllabus section of your course.

Question Type Grading Note: For multiple choice questions with multiple correct answers, you earn credit for each correct answer option selected; however, credit is automatically deducted for each incorrect answer option selected. You cannot earn less than zero (0) for any question. If no answer option is selected, you will earn zero (0) points for no attempt, so it is best to provide an answer to every question.	Multiple-choice questions with a single correct answer Multiple-choice questions with multiple correct answers Essay (free response) questions	Multiple-choice questions with a single correct answer Multiple-choice questions with multiple correct answers Essay (free response) questions
Number of Questions	24 total questions (23 content questions + 1 academic integrity question)	20 total questions (19 content questions + 1 academic integrity question)
Availability Start	Sunday, September 14, 2026, at 12:01 AM AZ Time	Sunday, October 4, 2026, at 12:01 AM AZ Time
Availability End	Sunday, September 20, 2026, at 11:59 PM AZ Time	Friday, October 9, 2026, at 11:59 PM AZ Time
Scheduling Reminder In order to have enough time to complete the exam, you should start your exam no later than the listed date and time to	Sunday, September 20, 2026, at 9:01 PM AZ Time	Friday, October 9, 2026, at 9:01 PM AZ Time

CSE 551 Syllabus
 Fall A 2026

Any information in this syllabus may be subject to change without advance notice. Stay informed by checking course announcements and the syllabus section of your course.

ensure you have enough time to complete it before the due date.		
Duration	120 minutes + plan for at least 15 minutes for proctoring set up	120 minutes + plan for at least 15 minutes for proctoring set up

Proctoring

Learners needing allowance accommodations need to work through the [Student Accessibility and Inclusive Learning Services \(SAILS\)](#) and/or the [Pat Tillman Veterans Center \(PTVC\)](#).

LockDown Browser

LockDown Browser will proctor your graded quizzes this session. It is an online proctoring service that allows you to take your exam from the comfort of your home. You do **not** need to create an account or schedule an appointment in advance. It is available twenty-four hours a day/seven days a week/and three hundred sixty-five days a year (24 hours a day/7 days a week/365 days a year), and all that is required is a computer, a working webcam/microphone, your ID, and a stable internet connection.

Review your onboarding course and *Module 0: Welcome and Start Here* in your course for more information about LockDown Browser and how to download the required software.

Graded Quiz Allowances

Any items **not** included in this list are **not** allowed during the assessment sessions or in your assessment space.

Reminders

- Instructors reserve the right to adjust scoring based on academic integrity violations (e.g., not following the allowances).

CSE 551 Syllabus
Fall A 2026

Any information in this syllabus may be subject to change without advance notice. Stay informed by checking course announcements and the syllabus section of your course.

- You are to independently take the assessment in a single session without leaving the testing space (e.g., no bathroom breaks) to ensure proctoring of the entire session.
- You will be allowed one (1) attempt to take and complete your assessment. Once you open your assessment, your testing session begins and you will need to complete it within the allotted time. Your assessment will automatically be submitted if it is **not** completed before the deadline.
- Select a location where you can comfortably take your assessment without interruption and have a video recording taken of yourself and your workspace environment. This area should also be free of distractions and interruptions.
- Turn off all other devices (e.g., tablets, phones, second computers) and place them outside of your reach.
- Clear your environment of all external materials **not** permitted. Review the allowances to ensure you know what is permitted.
- Before starting the assessment, know how much time is available for it by checking your course and syllabus, and that you have allotted sufficient time to complete it.
- Remain at your device for the duration of the assessment.
- If the device or networking environment is different than what was used previously with the **Webcam Check** and **System & Network Check** in LockDown Browser, run the checks again prior to starting the test.
- You are to stay within a clear view of the webcam throughout the duration of the proctored session.
- To produce a good webcam video:
 - Avoid wearing headwear (e.g., hats).
 - Ensure your device is on a firm surface (e.g., desk or table). Do **not** have the device on your lap, a bed or other surface where the device or you are likely to move.
 - If using a built-in webcam, avoid tilting the screen after the webcam setup is complete.

CSE 551 Syllabus

Fall A 2026

Any information in this syllabus may be subject to change without advance notice. Stay informed by checking course announcements and the syllabus section of your course.

- Take the assessment in a well-lit room and avoid backlighting, such as sitting with your back to an open window.
- Remember that LockDown Browser will prevent you from accessing other websites or applications; you will be unable to exit the assessment until all questions are completed and submitted unless otherwise directed in the allowances.
- Before the assessment concludes and the proctoring session ends, all scratch paper must be destroyed and all whiteboard markings must be erased.

Specific Allowances

- **Web links:** No
- **Open book:** No
- **Pre-written paper notes:** No
- **Scratch paper:** Yes
 - Unlimited amount of blank scratch paper of any size, writing utensils (e.g., pens, pencils, markers, and/or highlighters) and erasers; please have extra ones in your testing area should you run out of ink, the pencil breaks, etc.
 - Before the quiz concludes and the proctoring session ends, all scratch paper must be destroyed and all whiteboard markings must be erased. The last question in the quiz will be a confirmation of learners executing these ASU academic integrity actions.
- **On-Screen Calculator:** No
- **Handheld calculator:** No

Honorlock

Honorlock will proctor your exams this session. Honorlock is an online proctoring service. You do not need to create an account or schedule an appointment in advance. Honorlock is available twenty-four hours a day and seven days a week (24 hours/7 days).

CSE 551 Syllabus
Fall A 2026

Any information in this syllabus may be subject to change without advance notice. Stay informed by checking course announcements and the syllabus section of your course.

Honorlock will be enabled for at least one practice assessment in your course to prepare you for the proctored exams in this course.

Review your onboarding course and the Welcome and Start Here section in your course for more information about Honorlock and how to download the required Chrome Extension.

Exam Allowances

Any items not included in this list are **not** allowed during the exam or in your exam space.

Reminders

- You are to independently take your exams in a single session. Once you open your exam, your testing session begins and you will need to complete it within the allotted time. Your exam will automatically be submitted if it is not completed before the deadline. You will be allowed one (1) attempt to take and complete your exams.
- You are to independently take the exam in a single session without leaving the testing space (e.g., no bathroom breaks) to ensure proctoring of the entire session.
- You are to stay within a clear view of the webcam throughout the duration of the proctored exam session.
- You are required to set up a second, side-angle web camera for your proctored session. Refer to the Setting Up A Side-Angle Camera guide for directions on how to properly set up your device before beginning your session.
- Before the exam concludes and the proctoring session ends, all scratch paper must be destroyed and all whiteboard markings must be erased.
- The last question in the exam will be a confirmation of you upholding ASU academic integrity.

Specific Allowances

- **Site URLs:** No
- **Open book:** No
- **Pre-written paper notes:** Yes. Please read the specifications and plan ahead accordingly; you can only choose **one (1)** of these options:

CSE 551 Syllabus
Fall A 2026

Any information in this syllabus may be subject to change without advance notice. Stay informed by checking course announcements and the syllabus section of your course.

- **Handwritten notes:** Hard copy; two (2) sheets; standard letter (8.5 in x 11 in) or A-4 paper; double-sided.
- **Printed Notes:** Hard copy; one (1) sheet; standard letter (8.5 in x 11 in) or A-4 paper; double-sided.
- **Scratch paper:** Yes
 - Unlimited amount of blank scratch paper of any size, writing utensils (e.g., pens, pencils, markers, and/or highlighters) and erasers; please have extra ones in your testing area should you run out of ink, the pencil breaks, etc.
 - Before the exam concludes and the proctoring session ends, all scratch paper must be destroyed and all whiteboard markings must be erased. The last question in the exam will be a confirmation of learners executing these ASU academic integrity actions.
- **On-Screen Calculator:** No
- **Handheld calculator:** Yes
 - **Not** a separate device's calculator (e.g., mobile/cell phone) or anything with Internet access.
 - Scientific Calculator
- **Side-Angle Webcam:** No
 - External camera
 - External webcam
 - Mobile Device: Must run at least iOS 16+ or Android 7.1.1+
- **Restroom breaks:** No
- **Copy and Paste:** No
- **Hats:** No
- **Headphones:** No
- **Take Exam in a Public Area:** No
- **Mobile Phone Use:** No
- **Background Noise (Occasional sounds expected in the testing area):** No

Course Grade Breakdown

Course Work	Quantity	Team or Individual	Points
Homework Quizzes	8	Individual	20%
Module Quizzes	8	Individual	10%
Coding Assignments	8	Individual	10%
Exam 1	1	Individual	30%
Exam 2	1	Individual	30%
Total Course Points			100%

This course is **not** portfolio eligible because there are no course projects accounting for 30% or more of the overall course grade.

Grade Scale

Being a very challenging theory course and letter grades being based on two (2) exams, the grade scale serves as an **estimate** for letter grades. This is why the point values have such large ranges. A traditional curve will **not** be used to assign letter grades because learners will **not** be competing against one another; however, the instructor reserves the right to categorize grades, which is often to the benefit of the learners and is based on academic performance.

You must earn at least a "C" to receive graduate credit. This course has **no** grade curving. **All** graded coursework will be included to calculate grades (i.e., no graded items will be dropped). Grades will **not** be rounded.

The instructor reserves the right to adjust individual grades, including possible pluses and minuses, based on, but not limited to: violations of academic integrity, achievement groupings, etc.

CSE 551 Syllabus
Fall A 2026

Any information in this syllabus may be subject to change without advance notice. Stay informed by checking course announcements and the syllabus section of your course.

Percentage	Letter Grade
90 - 100	A+
80 - 89.99	A
70 - 79.99	B+
60 - 69.99	B
50 - 59.99	C
40 - 40.99	D
0 - 39.99	E

Zoom Meetings

This course may have three (3) types of Zoom meeting opportunities:

- Instructor Zoom Events:** If learners have specific questions or topics of interest to be discussed, they can indicate their request in an Ed Discussion thread prior to the meeting. Although it may not be possible to address all requests during the Zoom event, the instructor is interested in tailoring this time to the learners' questions and interests. The instructor will be following a set agenda, so please be mindful of that when engaging in the Zoom event.
- Instructional Assistant (IA) Zoom Support Sessions:** These sessions offer a chance for learners to get their questions answered from the IA(s). Although the course team is responsive to trends in Ed Discussion, these events focus on addressing learners' specific questions related to content: clarifications, reteaching, assessment preparation, etc.

CSE 551 Syllabus
 Fall A 2026

Any information in this syllabus may be subject to change without advance notice. Stay informed by checking course announcements and the syllabus section of your course.

- **Grader Zoom Support Sessions:** These sessions are singularly focused on addressing questions related to **grading**.

Check the Zoom tab in the navigation menu of your course. Although we try to be consistent for our learners' planning purposes, the schedule is subject to change throughout the course, so stay up-to-date on the event details by checking Ed Discussion and course announcements.

Read about the specific policies related to Zoom meetings directly in your onboarding course and your course pages: Syllabus, ASU Course Policies, and any additional course-specific policy information in *Module 0: Welcome and Start Here*. Additional information may be included in the Policies section of this syllabus. You are responsible for adhering to all policies.

Zoom Recordings

- Instructor Zoom Events are **recorded and shared** through the "Zoom" navigation link in your course. These can be found by going to the "Cloud Recordings" tab. These recordings will be unavailable after 120 days.
- IA Zoom Support Sessions are **recorded**, but **not uploaded** into the course. It is at the discretion of the instructor if these sessions will be shared during the course session.
- Grader Zoom Support Sessions are **recorded**, but **not uploaded** into the course.

Policies

Please refer to the ASU Course Policies section in your course, your onboarding course, and *Module 0: Welcome and Start Here* of your course in addition to the policies listed in this section.

Code of Conduct

The aim of education is the intellectual, personal, social, and ethical development of the individual. The educational process is ideally conducted in an environment that encourages reasoned discourse, intellectual honesty, openness to constructive change, and respect for the rights of all individuals. Self-discipline and a respect for the rights of others in the university community are necessary for the fulfillment of such goals. The [Student Code of Conduct](#) is designed to promote this environment at Arizona State University.

CSE 551 Syllabus
Fall A 2026

Any information in this syllabus may be subject to change without advance notice. Stay informed by checking course announcements and the syllabus section of your course.

The [Student Code of Conduct](#) sets forth the standards of conduct expected of students who choose to join the university community. Students who violate these standards will be subject to disciplinary sanctions in order to promote their own personal development, to protect the university community, and to maintain order and stability on campus.

Students are expected to follow the [ABOR Student Code of Conduct](#) and the ASU [Student Code of Conduct](#) at all times.

Expected Behavior

Students are expected to follow the ASU [Student Code of Conduct](#) and are expected to acknowledge and embrace the [FSE Student Professionalism Expectation](#).

An instructor may withdraw a student from a course with a mark of "W" or "E" or employ other interventions when the student's behavior disrupts the educational process. For more information, review [SSM 201-10](#).

If you identify something as unacceptable classroom behavior (e.g., in Canvas, Ed Discussion, Zoom, etc.), please notify the course team.

Our classroom community rules are to:

- Be professional
- Be positive
- Be polite
- Be proactive

Academic Integrity

All engineering students are expected to adhere to the ASU Student [Honor Code](#) and the ASU academic integrity policy, which can be found at <https://provost.asu.edu/academic-integrity/policy>. Students are responsible for reviewing this policy and understanding each of the areas in which academic dishonesty can occur. If you have taken this course before, you may not reuse or submit any part of your previous assignments without the express written permission from the instructor.

All student academic integrity violations are reported to the Fulton Schools of Engineering Academic Integrity Office (AIO). Withdrawing from this course will not absolve you of responsibility for an

CSE 551 Syllabus
Fall A 2026

Any information in this syllabus may be subject to change without advance notice. Stay informed by checking course announcements and the syllabus section of your course.

academic integrity violation and any sanctions that are applied. The AIO maintains a record of all violations and has access to academic integrity violations committed in all other ASU college/schools.

Honorlock Proctoring

This course will have side-angle camera Honorlock proctored exams. For all Honorlock proctored exams, you must have an external web cam that can be set up so that there is a side-angle view of your entire testing area. This camera must be an external camera and may not be the embedded camera on the computer in which you are taking the exam. While a corded external camera is preferred, Android phones running at least 7.1.1+ or iPhones running iOS 16+ may be used as an external camera. Refer to the Honorlock Mobile Application - Test Taker Experience guide for details on using a smartphone with Honorlock.

No Generative AI Use Permitted

In this course, all assignments must be completed by the student. Artificial Intelligence (AI), including ChatGPT and other related tools used for creating of text, images, computer code, audio, or other media, are not permitted for use in any work in this class. Use of these generative AI tools will be considered a violation of the [ASU Academic Integrity Policy](#), and students may be sanctioned for confirmed, non-allowable use in this course.

Copyright

You must refrain from uploading to any course shell, discussion board, or website used by the course instructor or other course forum, material that is not the student's/learner's original work, unless the student/learner first complies with all applicable copyright laws; faculty members reserve the right to delete materials on the grounds of suspected copyright infringement.

The contents of this course, including lectures (Zoom recorded lectures included) and other instructional materials, are copyrighted materials. Students may not share outside the class, including uploading, selling or distributing course content or notes taken during the conduct of the course. Any recording of class sessions is authorized only for the use of students enrolled in this course during their enrollment in this course. Recordings and excerpts of recordings may not be distributed to others (see [ACD 304-06](#), "Commercial Note Taking Services" and [ABOR Policy 5-308 F.14](#) for more information).

CSE 551 Syllabus Fall A 2026

Any information in this syllabus may be subject to change without advance notice. Stay informed by checking course announcements and the syllabus section of your course.

Graded Quiz and Exam Policy

Each course in the MCS program is uniquely designed by expert faculty so that learners can best master the learning outcomes specific to each course. By design, course features and experiences are different across all MCS courses.

In the MCS program, we strive to provide learners with exercises and applied practice beyond quizzes and exams that align with the hands-on nature of the computer science industry. Ungraded practice opportunities may include, but are not limited to: in-video-questions (IVQs), knowledge check quizzes (KCs), module practice quizzes, practice exams, and other coursework. When available, the questions and correct answers are provided to learners. Depending on the type of questions, auto-generated feedback is built into the course to further help learners learn in real-time. Please thoroughly review your course to ensure that you are aware of the types of practice opportunities available to you.

For academic integrity purposes, once grades are made available, learners will see their overall total scores. Like other standardized tests, such as the GRE and SAT, learners will receive a singular grade for the exams, but the questions, correct and incorrect answers, and feedback to each question will **not** be provided.

If learners desire 1:1 feedback, please send a private thread to the course team on Ed and/or attend a Zoom meeting with the course team. Rather than receiving the exact questions learners had correct and incorrect and the answers to those questions, learners will likely receive the concepts that were covered in the assessment questions so they will know what they need to review prior to other assessments and how to apply this information in their professional environments.

Absence Policies

There are no required or mandatory attendance events in this online course. Different types of Zoom meetings hosted by any course team member do not take attendance.

Excused absences do not relieve students of responsibility for any part of the coursework required during the period of absence. If exceptions for graded coursework deadlines need to be made for known excused absences, please reach out to the course team by the end of the second week of the course by sending a private thread to the course team on Ed. Review availability windows and due dates for coursework and schedule accordingly. The exam availability windows allow for your own flexibility and you are expected to plan ahead.

CSE 551 Syllabus
Fall A 2026

Any information in this syllabus may be subject to change without advance notice. Stay informed by checking course announcements and the syllabus section of your course.

Review the resources for what qualifies as an excused absence and review the late penalties in the Assignment Deadlines and Late Penalties section of the syllabus and the course:

- Excused absences related to religious observances/practices that are in accord with [ACD 304-04](#), “Accommodation for Religious Practices” (please see [Religious Holidays and Observances](#)).
- Excused absences related to university sanctioned events/activities that are in accord with [ACD 304-02](#), “Missed Classes Due to University-Sanctioned Activities”.
- Excused absences related to missed class due to military line-of-duty activities that are in accord with [ACD 304-11](#), “Missed Class Due to Military Line-of-Duty Activities,” and [SSM 201-18](#), “Accommodating Active Duty Military”.

Zoom Meetings

Recording Notice

Instructors will record Zoom meetings. An archived recording will be made available in Canvas for enrolled students, instructors, or support personnel. Creation of recordings for individuals or groups beyond these requires consent from students who are recorded.

Expectations

The environment should remain professional at all times. Inappropriate content/visuals, language, tone, feedback, etc. will not be tolerated, reported and subject to disciplinary action. Review the policy regarding Expected Behavior section of the syllabus, ASU [Student Code of Conduct](#), and [FSE Student Professionalism Expectation](#) for more detailed information.

Policy Against Threatening Behavior, per the Student Services Manual, ([SSM 104-02](#))

Students, faculty, staff, and other individuals do not have an unqualified right of access to university grounds, property, or services (see [SSM 104-02](#)). Interfering with the peaceful conduct of university-related business or activities or remaining on campus grounds after a request to leave may be considered a crime. All incidents and allegations of violent or threatening conduct by an ASU

CSE 551 Syllabus

Fall A 2026

Any information in this syllabus may be subject to change without advance notice. Stay informed by checking course announcements and the syllabus section of your course.

student (whether on- or off-campus) must be reported to the ASU Police Department (ASU PD) and the Office of the Dean of Students.

Disability Accommodations

Suitable accommodations are made for students with disabilities. Students needing accommodations must register with [ASU Student Accessibility and Inclusive Learning Services](#) (SAILS) office. Students should communicate the need for an accommodation at the beginning of each course so there is sufficient time for it to be properly arranged. These requests should be submitted through the [online portal](#). See [ACD 304-08](#) Classroom and Testing Accommodations for Students with Disabilities. SAILS will send the instructor a notification of accommodations. It is recommended that students communicate with instructors regarding documented accommodations.

Harassment and Sexual Discrimination

Arizona State University is committed to providing an environment free of discrimination, harassment, or retaliation for the entire university community, including all students, faculty members, staff employees, and guests. ASU expressly prohibits discrimination, harassment, and retaliation by employees, students, contractors, or agents of the university based on any protected status: race, color, religion, sex, national origin, age, disability, veteran status, sexual orientation, gender identity, and genetic information.

Title IX is a federal law that provides that no person be excluded on the basis of sex from participation in, be denied benefits of, or be subjected to discrimination under any education program or activity. Both Title IX and university policy make clear that sexual violence and harassment based on sex is prohibited. An individual who believes they have been subjected to sexual violence or harassed on the basis of sex can seek support, including counseling and academic support, from the university. If you or someone you know has been harassed on the basis of sex or sexually assaulted, you can find information and resources at <https://sexualviolenceprevention.asu.edu/faqs>.

As a mandated reporter, I am obligated to report any information I become aware of regarding alleged acts of sexual discrimination, including sexual violence and dating violence. ASU Counseling Services, <https://eoss.asu.edu/counseling> is available if you wish to discuss any concerns confidentially and privately. ASU online students may access 360 Life Services, <https://goto.asuonline.asu.edu/success/online-resources.html>.

CSE 551 Syllabus
Fall A 2026

Any information in this syllabus may be subject to change without advance notice. Stay informed by checking course announcements and the syllabus section of your course.

Photo Requirement

Arizona State University requires each enrolled student and university employee to have on file with ASU a current photo that meets ASU's requirements (your "Photo"). ASU uses your Photo to identify you, as necessary, to provide you educational and related services as an enrolled student at ASU. If you do not have an acceptable Photo on file with ASU, or if you do not consent to the use of your photo, access to ASU resources, including access to course material or grades (online or in person) may be negatively affected, withheld or denied.

Warning of Offensive Class Materials

If you feel any course material is offensive, please reach out to the instructor by sending a private thread on Ed.

Course Creator



Andréa Richa, PhD

Andréa Richa, Ph.D. joined Arizona State University (ASU) in 1998. She is currently affiliated with the Biomimicry Center at ASU, and the Biosocial Complexity Initiative in general. Prof. Richa's main areas of expertise are in distributed/network algorithms and computing in general. More recently she has focused on developing the algorithmic foundations on what has been coined as programmable matter, through her work on self-organizing particle systems (SOPS) (see sops.engineering.asu.edu). Her work has been widely cited, and includes, besides SOPS, work on bio-inspired distributed algorithms, distributed load balancing, packet routing, wireless network modeling and topology control, wireless jamming, data mule networks, underwater optical networking, and

CSE 551 Syllabus Fall A 2026

Any information in this syllabus may be subject to change without advance notice. Stay informed by checking course announcements and the syllabus section of your course.

distributed hash tables (DHTs). Dr. Richa received the 2017 Best Senior Researcher award from the School of Computing, Informatics, and Decision Systems Engineering (CIDSE). She was the recipient of an NSF CAREER Award in 1999, an Associate Editor of IEEE Transactions on Mobile Computing, and the keynote speaker and program\general chair of several prestigious conferences. In particular, Prof. Richa was the Program Committee Chair of the 31st International Symposium on Distributed Computing (DISC), 2017, one of the top two conferences in distributed computing. Prof. Richa has also delivered several invited talks both nationally and internationally. For a selected list of her publications and other accomplishments, CV, and current research projects, please visit www.public.asu.edu/~aricha or sops.engineering.asu.edu.

CSE 551 Syllabus

Fall A 2026

Any information in this syllabus may be subject to change without advance notice. Stay informed by checking course announcements and the syllabus section of your course.